EA Credit: Grid Harmonization

This prerequisite applies to

- BD+C: New Construction (1-2 points)
- BD+C: Core & Shell (1-2 points)
- BD+C: Schools (1-2 points)
- BD+C: Retail (1-2 points)
- BD+C: Data Centers (1-2 points)
- BD+C: Warehouses & Distribution Centers (1-2 points)
- BD+C: Hospitality (1-2 points)
- BD+C: Healthcare (1-2 points)

Intent

To increase participation in demand response technologies and programs that make energy generation and distribution systems more efficient, increase grid reliability, and reduce greenhouse gas emissions.

Requirements

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Design building and equipment for participation in demand response programs through load shedding or shifting. On-site electricity generation does not meet the intent of this credit.

Case 1. Demand Response Program Available and Participation (2 points)

- Participate in an existing demand response (DR) program and complete the following activities. Design a system with the capability for real-time, fully-automated DR based on external initiation by a DR Program Provider. Semi-automated DR may be utilized in practice.
- Enroll in a minimum one-year DR participation amount contractual commitment with a qualified DR program provider, with the intention of multiyear renewal, for at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance. The on-peak demand may vary based on the utility climate and pricing structures.
- Develop a comprehensive plan for meeting the contractual commitment during a Demand Response event.
- Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.
- Include the DR program and any installed technologies in the building systems manual.
- Initiate at least one full test of the DR plan.

OR

Case 2. Demand Response Capable Building (1 point)

Have infrastructure in place to take advantage of future demand response programs or dynamic, real-time pricing programs and complete the following activities:

- Install interval recording meters with communications and ability for the building automation system to accept an external price or control signal.
- Develop a comprehensive plan for shedding at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance.
- Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.
- Include the DR program and any installed technologies in the building systems manual.
- Contact local utility representatives to discuss participation in future DR programs.

AND / OR

Case 3. Load Flexibility and Management Strategies (1-2 points)

Analyze the building's annual load shape and peak load based as calculated for EA prerequisite Minimum Energy Performance. Review the regional grid load profile using the metric of peak load or peak carbon emissions. The U.S. Environmental Protection Agency's (EPA) AVoided Emissions and geneRation Tool (AVERT) provides regional grid emissions data; local utilities may also provide this data.

Coordinate review of building load shape and peak load with review of the regional grid profile to identify the best value load management strategies that the building can provide.

Implement one or more of the load flexibility and management strategies described below for a maximum of up to two points. All projects must install interval recording meters with communications and the ability for the building automation system to accept an external price signal.

Load Flexibility and Management Strategies:

- Peak Load Optimization: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point)
- Flexible Operating Scenarios: demonstrate that strategy moves at least 10% of peak load by a time period of 2 hours (1 point)
- On-site thermal and/or electricity storage: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point)
- Grid resilience technologies: project served by utilities with resilience programs in place, which leverage strategies such as islanding and part-load operation, automatically achieve this credit (1 point)

Include installed technology in the scope of work for the commissioning authority. Include load flexibility and management strategies and installed technologies in the building systems manual.

Contact local utility representatives to discuss participation in future DR programs and to inform utility of building load flexibility and management strategies.

Guidance

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

As the number of distributed energy resources, or grid-connected power generation from individual buildings, increases, the utility must integrate these resources while continuing to manage overall grid capacity and generation. Too much distributed generation that is not integrated into the energy system can impair grid operation. Buildings have the opportunity to support effective grid management by designing on-site electricity systems that are integrated components of the energy system, serving as resources for the grid rather than disruptors.

In addition to participation in a utility's demand response program, projects can install storage technologies and implement operational strategies that support effective grid management and increase grid reliability. Eligible technologies may include battery storage, flywheel energy storage, or thermal energy storage; load management strategies may include load shifting or flexible operating scenarios.

Beta Update

Updates intend to address projects where no demand response program is available from the local utility and recognize new distributed energy resources technologies and grid harmonization strategies.

Step-by-Step Guidance

Refer to Steps 1 – 5 in the LEED v4 reference guide, with the following addition:

Step 6. Include demand response in commissioning and in building systems manual

Coordinate with the CxA to include a review of the DR plan in the commissioning of the building's system test procedures, to verify the ability to handle an externally initiated demand response event. The Cx plan must include at least one performance test of the full DR plan to verify that all equipment responds as planned and that all responsible parties understand their roles.

Include demand response program and any installed technologies in the building systems manual.

Case 1. Demand Response Program Available

Refer to the LEED v4 reference guide, with the following addition:

Step 2. Initiate at least one full test of the DR plan.

Execute at least one performance test of the full DR plan as defined in the Cx plan. All relevant DR operations team members must participate, and address and mitigate any issues identified as a result of the performance test.

Case 2. Demand Response Program Not Available

Refer to the LEED v4 reference guide.

Case 3. Load Flexibility and Management Strategies

Step 1. Analyze building load shape

Analyze building load shape as calculated for EA prerequisite Minimum Energy Performance; compare building load shape with the regional grid profile using the metric of peak load or peak carbon emissions. Use the U.S. Environmental Protection Agency's AVoided Emissions and geneRation Tool (AVERT) provides regional grid emissions data, or contact the local utility to request this data.

Step 2. Identify highest-value strategies

Based on the analysis of building load shape, peak load and regional grid peak load or peak carbon emissions, identify storage technologies or load management strategies that the building can implement at the lowest cost while providing the greatest value to the grid.

Step 3. Implement strategies

Implement one or more of the strategies identified during step 2.

Eligible strategies include peak load optimization, or reducing on-peak load by at least 10% compared to peak electrical demand, or flexible operating scenarios, where a building moves at least 10% of its peak load by a time period of 2 hours. Additionally, the project may install on-site thermal energy storage for heating and cooling and/or electricity storage capable of reducing on-peak load by at least 10% as compared to peak electrical demand. On-site storage enables the building to store energy and use it during peak demand times, increasing annual energy savings and reducing strain on the grid.

If a project is served by a utility with a resilience program in place that leverages strategies such as islanding or part-load operation, the project can automatically earn 1 point under this credit by providing documentation of the utility resilience program in place.

Step 4. Include load flexibility and management system in commissioning and in building systems manual

Coordinate with the CxA to include a review of the load management system and supporting technologies in the commissioning of the building's system test procedures. The Cx plan must include at least one performance test of the full system to verify that all equipment operates as planned and that all responsible parties understand their roles.

Include the system and any installed technologies in the building systems manual.

Step 5. Contact local utility

Contact the local utility or service provider to express interest in participation in a future demand response program, and to inform the utility of building load flexibility and management strategies.

Further Explanation
Grid Harmonization

Energy efficient and grid-interactive buildings save money and resources while supporting broader grid-scale decarbonization. Savings to building owners accrue from a combination of demand charge reductions, lower annual energy use, utility incentives, and increase building resilience.

Required Documentation

Documentation	Case 1	Case 2	Case 3
Proof of enrollment in DR program	X		
Evidence of ability to shed 10% of on-peak demand	X	X	X
Confirmation that system is capable of receiving and acting on external signal	X	X	X
Action plan for meeting reduction requirement during event	X	X	X
Inclusion of DR in CxA systems testing plan	X	X	X
Inclusion of DR and/or grid harmonization technologies in building systems manual	X	X	X
Documentation of one full test of the DR plan	X		
Narrative or report that includes: summary of building annual load shape and regional grid profile analysis; description of building load flexibility and/or management strategies implemented			X
Documentation of grid resilience program or technologies serving the project, as applicable			X

Connection to Ongoing Building Performance

- Participation in a demand response program enables projects to support efficient energy generation and distribution systems, increase grid reliability, and reduce greenhouse gas emissions. For projects where no demand response program is available, implementing load flexibility and management strategies can help to achieve the same outcomes and support effective electrical grid management.